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Compressed Sensing

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Published	IEEE Transactions on Information Theory 52(4):1289-1306 (2006)
DOI	https://doi.org/10.1109/TIT.2006.871582
Paper Link	https://doi.org/10.1109/TIT.2006.871582
Domain	Information Theory / Compressed Sensing

Abstract (verbatim)

Suppose x is an unknown vector in R^m (a digital image or discrete signal); we plan to measure n general linear and nonadaptive measurements of x with n much smaller than m. Can we reconstruct x accurately from these incomplete observations? We prove that if x has a sparse representation in some orthonormal basis, it can be reconstructed exactly from very few measurements by solving an l1-norm optimization problem. This establishes that unobserved high-dimensional information can be perfectly recovered under sparsity constraints.

Donoho (2006), IEEE Trans. Inf. Theory

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